B

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Introduction

The goal of this experiment was to identify an unknown compound through examining three types of spectral data. Using data collected from three analytical methods, NMR, FTIR, and MS, the unknown "B" was identified. These scans show different information about the makeup of the unknown and in using data from all three scans, the structure of the unknown can be determined. This analysis can be very useful when a new compound has been synthesized and its structure needs to be identified.

Body

1. NMR

The Nuclear Magnetic Resonance (NMR) method for analysis is one of many analytical techniques used by laboratory scientists to identify organic compounds. The way NMR works is by measuring the transitions of magnetic energy levels within protons which occurs when atomic nuclei are immersed into an external magnetic field and bombarded with electromagnetic radiation. Protons possess a physical property known as spin and through the rotation of these charged protons, a magnetic field is generated. This carries a magnetic moment, which measures an object's tendency to align with a magnetic field. In an NMR experiment, a sample of a substance is placed into a metal coil and is exposed to a magnetic field. This forces the nucleus' magnetic moments to align parallel to the applied field. Then the sample is polarized or magnetized. Then the sample is irradiated with a broad spectrum of resonance frequencies and each of the various protons in a molecule should respond to a specific resonance frequency. This provides information about the position of atoms present in the molecule. A receiver coil records a complex signal, or free induction decay (FID), which is all of the electrical impulses generated by each type of proton. Then FID is converted into a spectrum through a mathematical technique

called Fourier transform which can unravel all the individual frequencies. This allows the NMR signal to be displayed on a spectrum, where the characteristic frequencies of the processing nuclear magnetic moments are shown as distinct peaks called resonance lines. From here the molecular structure can be determined based on the location and magnitude of the NMR spectrum.

2. FTIR

Fourier-Transform Infrared Spectroscopy (FTIR) is the analysis of infrared light interacting with a molecule by measuring the emission, absorption, and reflection in order to determine functional groups of molecules. Molecules with different structures and bonds do not absorb the same energies of infrared radiation and will give different absorption patterns. Once absorbed, the IR waves affect vibrational modes of molecules. Molecules are capable of being bent in different ways and each movement will result in different absorption signals on the IR spectrum. A plot is constructed showing which frequencies were absorbed by the sample. The most used type of spectrometer, a Fourier-transform spectrometer, irradiates the sample with each frequency simultaneously and utilizes a mathematical operation called a Fourier transform to determine the frequencies that passed through the sample. These frequencies are then converted into a wavenumber and plotted against their relative transmittance. This graph's intensity, wavenumber, and shape can be used to determine the functional groups and their location in a compound.

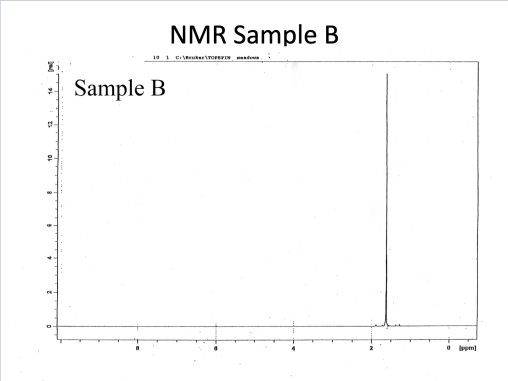
3. MS

Mass Spectrometry (MS) is the study of the interaction between matter and an energy source other than electromagnetic radiation. MS is primarily used in order to determine the formula and molecular weight of a compound. In a mass spectrometer, a compound is first

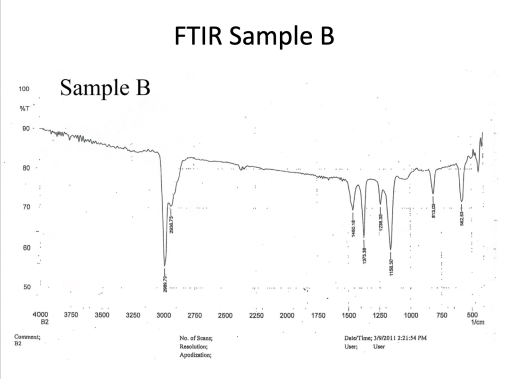
vaporized and converted into ions which are then separated and detected. The most common ionization technique involves bombarding the compound with high-energy electrons which strike the molecule and causes one of the electrons in the molecule to be ejected. This technique is called electron impact ionization (EI) and it generates a high-energy intermediate that is a radical cation which is called the parent ion. The parent ion is often very unstable and susceptible to fragmentation, which generates fragmented ions from the original compound. The molecular ion and all the fragments can be analyzed according to mass to charge ratio and their relative abundance to yield structural information about the sample.

Data

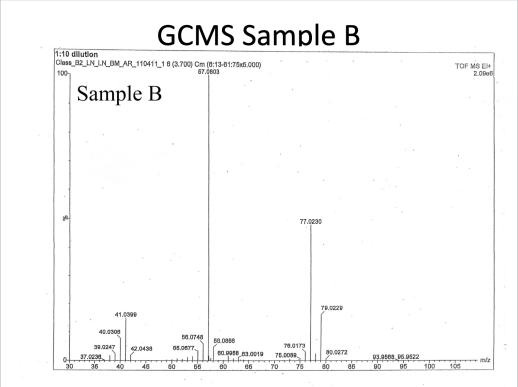
1. NMR



2. FTIR



3. MS



Analysis

1. NMR

In the NMR graph there is a peak at approximately 1.6 ppm. Singular peaks indicate that all of the protons in a compound are magnetically equivalent. There is also one small peak to the left and two small peaks to the right of the main one. This indicates that there are three methyl groups.

2. FTIR

Since the FTIR data had no peaks between 1600-2700, there are no double or triple bonds. Since there is an intense peak at 2989, there are also multiple carbon hydrogen bonds. There are no other peaks in this region, so it can be inferred that the only bonds to hydrogen are carbon.

3. MS

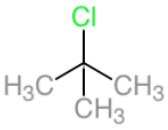
The MS data shows a parent peak with an m/z of 94 indicating that the molar mass of the unknown is also 94. The base peak has an m/z of 57, which shows fragmentation of Chlorine-37 since 94-57=37. The base peak also indicates a tert-butyl group with a mass of 57. Chlorine's

presence is also indicated by the peaks at 77 and 79, since peak 77 is at about 50% abundance and peak 79 is at approximately 15% abundance. These two peaks represent Cl^{35} and Cl^{37} respectively. Cl^{35} is about three times more abundant than Cl^{37}, so their abundance in the compound is also 3:1. This can be seen through the peak's intensity.

Conclusion

Each set of data showed results indicating the same compound. A t-butyl group connected to a carbon atom has only one kind of proton and the intensity in the areas of electron density of the protons were the same. Also, the beta carbon for each proton was the central carbon connected chlorine, and this carbon had no protons and therefore matches with NMR's result of the lack of splitting. Due to this, our unknown is determined to be t-Butyl chloride.

The unknown "B" has been identified as 2-Chloro-2-methylpropane.



References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Laboratory Manual [lab manual]. Auburn (AL): Auburn University.